

Strong Years

12-Week Printable



Your wall plan for four levels: Rebuild, Steady, Strong and Iron.
Monthly Strength Age retest sheet and habit tracker.

From Chang Yin, for your fridge.

Chang Yin and Sun Yoon are AI characters created by a team of people. Their life story is fiction. The exercises and recipes are built on published research for older adults (sources in the back). This is education, not medical advice. Check with your doctor before starting new exercise.

How to use your 12-week plan

Chang Yin

I'm Chang. AI character. This goes on your fridge, not in a drawer.

Pick your track. Do Monday and Thursday strength every week. The other days are shorter. Every four weeks, you retest and write the number down. That's the whole plan. Strong is a habit.

Pick your track. Rebuild if you need your hands to get out of a chair or use a walker. Steady if you can stand up without your hands but don't exercise. Strong if Steady is easy. Iron only if you already train. Not sure? Start one track lower.

Why twice a week for strength. A review of 121 trials in 6,700 older adults training 2 to 3 times a week found large gains in strength and easier chair stands (E01). The NSCA recommends 2 to 3 sessions a week, 1 to 3 sets of 6 to 12 reps, progressing gradually (E02).

How it progresses. Every two weeks the dose goes up a little: a rep, a set, a slower lowering, a heavier jug. Weeks 4, 8 and 12 end with your Strength Age retest.

Safety, every day: Stop if you feel chest pain or pressure, dizziness, unusual shortness of breath, or sharp pain. Chest pain, face drooping, arm weakness or trouble speaking? Call 911 now. Pain rule: Mild discomfort up to 3 out of 10 is okay if it settles by tomorrow. Sharper, or worse the next day: back off and ask your doctor or physical therapist. Chair against a wall, counter within reach, shoes on, no rugs. Sit a moment before you stand up from the floor or bed.

Rebuild track: your 12-week wall plan

Chair-based. Seated, or standing with both hands on the counter and the chair right behind you.

Mon strength · Tue mobility · Wed balance · Thu strength · Fri breath · Sat walk · Sun rest. ★ = retest week.

▲ = advanced. Move up when every rep felt easy two sessions in a row; move down any day.

Move	Weeks 1–2	Weeks 3–4 ★	Weeks 5–6	Weeks 7–8 ★	Weeks 9–10	Weeks 11–12 ★
Chair sit-to-stand (Mon, Thu)	2×5, hands on thighs, cushion on seat	2×6, hands on thighs	3×6	3×8	3×8, no cushion	3×8, arms crossed if you can
Seated chest press (Mon, Thu)	2×8, empty hands	2×10, water bottles	3×10	3×12	3×10, band behind chair	3×12, band
Seated towel pull-apart (Mon, Thu)	3 pulls, 3-s squeeze	4 pulls, 3 s	5 pulls, 3 s	5 pulls, 5 s	3×10 band pulls	3×12 band pulls
Seated heel raises (Mon, Wed, Thu)	2×10	2×12	3×12	3×12, 2-s pause at the top	Standing, both hands on counter, 2×10	Standing, 3×10
Seated knee straightening (Mon, Thu)	1×8 each leg	2×8 each	2×10 each	3×10 each	3×10, 3-s hold	3×10, 5-s hold
Balance at the counter, both hands (Wed + daily)	Feet together 3×10 s	Feet together 3×20 s	Half-step 3×10 s	Half-step 3×20 s	Heel-to-toe 3×10 s	Heel-to-toe 3×15 s
Walking (most days)	3–5 min indoors, twice a day	5 min, twice a day	7 min, twice	10 min, twice	10–15 min, twice	15 min, twice

Steady track: your 12-week wall plan

Standard versions with counter or wall support. Most new members start here. Mon strength · Tue mobility · Wed balance · Thu strength · Fri breath · Sat walk · Sun rest. ★ = retest week. ▲ = advanced.

Move up when every rep felt easy two sessions in a row; move down any day.

Move	Weeks 1–2	Weeks 3–4 ★	Weeks 5–6	Weeks 7–8 ★	Weeks 9–10	Weeks 11–12 ★
Sit-to-stand, arms crossed (Mon, Thu)	2×8	3×8	3×10	3×10, 3-s lower	3×12, 3-s lower	3×8 power stands (fast up, slow down)
Push (Mon, Thu)	Wall push-up 2×8	Wall 3×10	Wall 3×12, feet back	Counter push-up 2×8	Counter 3×8	Counter 3×10
One-arm jug row (Mon, Thu)	Half-gallon 2×8 each	Half-gallon 3×10	Gallon 2×8	Gallon 3×8	Gallon 3×10	Gallon 3×10, 1-s pause
Heel raises at the counter (Mon, Thu)	2×10	3×10	3×12	One hand 3×10	One hand 3×12	Single leg 2×6 each, both hands
Step-ups, hand on rail (Thu)	Toe taps 2×10	Toe taps 3×10	Step-ups 2×6 each	3×6 each	3×8 each	3×10 each
High wall sit (Mon)	2×15 s	2×20 s	3×20 s	3×20 s	3×25 s	3×30 s
Balance at the counter (Wed + toothbrush daily)	Heel-to-toe 3×10 s, hands on	Heel-to-toe 3×15 s	One leg 3×10 s, hands on	One leg 3×15 s	Heel-to-toe, hands hovering 3×15 s	One leg, one hand hovering 3×15 s
Walking (most days)	15 min	20 min	20 min + 2×1 min brisk	25 min + 3×1 min brisk	30 min + 3×90 s brisk	30 min + 4×90 s brisk

Strong track: your 12-week wall plan

Added load, slower lowering, less hand support. For people who can do Steady comfortably. Mon strength · Tue mobility · Wed balance · Thu strength · Fri breath · Sat walk · Sun rest. ★ = retest week. ▲ = advanced. Move up when every rep felt easy two sessions in a row; move down any day.

Move	Weeks 1–2	Weeks 3–4 ★	Weeks 5–6	Weeks 7–8 ★	Weeks 9–10	Weeks 11–12 ★
Sit-to-stand (Mon, Thu)	3×8, 3-s lower	3×10, 3-s lower	3×8, jug at chest	3×10, jug at chest	3×6 power stands	3×6 power stands, jug
Push (Mon, Thu)	Wall 2×10, feet back	Counter 2×8	Counter 3×8	Counter 3×10	Counter 3×10, 3-s lower	Counter 3×12, 3-s lower
Jug row (Mon, Thu)	Gallon 2×10 each	Gallon 3×10	Gallon 3×10, pause	Heavier bag 3×8	Heavier bag 3×10	Heavier bag 3×12
Step-ups (Thu)	2×6 each	3×6 each	3×8 each	3×8, bag in free hand	3×10, bag	3×10, heavier bag
Hinge pick-up (Thu)	Jug from chair 2×8	3×8	3×10	Jug from bottom step 3×8	3×10	Heavier, from step 3×10
Grocery carry (Thu)	2×30 s, two bags	3×30 s	3×40 s	3×40 s, heavier	3×45 s	3×60 s
Balance (Wed + daily)	Heel-to-toe, one hand hovering 3×15 s	3×20 s	One leg, one hand hovering 3×15 s	3×20 s	Backward walk, hand on counter, 2 lengths	One leg + clock reach 3 each
Walking (most days)	30 min	30 min + 3×90 s brisk	35 min + 4×90 s	40 min + 4×2 min	40 min, hills	45 min, hills

Iron track: your 12-week wall plan

For people who already train. Anything marked advanced is never your starting point. Mon strength

· Tue mobility · Wed balance · Thu strength · Fri breath · Sat walk · Sun rest. ★ = retest week. ▲ =

advanced. Move up when every rep felt easy two sessions in a row; move down any day.

Move	Weeks 1-2	Weeks 3-4 ★	Weeks 5-6	Weeks 7-8 ★	Weeks 9-10	Weeks 11-12 ★
Legs (Mon, Thu)	Sit-to-stand, jug, 3×10	Counter split squat 3×8 each	Split squat 3×10	Split squat, jug, 3×8 ▲	Split squat, jug, 3×10 ▲	Power stands, jug, 4×6 ▲
Push (Mon, Thu)	Counter 3×10, 3-s lower	Counter 3×12	Counter 3×12, 3-s lower	Bottom-stair push-up 3×6 ▲	Stair 3×8 ▲	Stair 3×10 ▲
Row (Mon, Thu)	Gallon 3×12, pause	Heavy bag 3×10	Heavy bag 3×12	Kettlebell or heavy bag 4×10 ▲	4×12 ▲	4×12, 2-s pause ▲
Step-ups (Thu)	3×8, bag	3×10, bag	3×10, heavier bag	Higher step 3×8	Higher step 3×10	Higher step, bag, 3×10 ▲
Hinge pick-up (Thu)	From step 3×8	3×10	Heavier 3×8	Heavier 3×10	4×8 ▲	4×10 ▲
Carry (Thu)	3×45 s heavy	3×60 s	4×45 s heavier	4×60 s ▲	4×60 s, one hand, switch ▲	5×60 s ▲
Balance (Wed + daily)	One leg, hands hovering 3×30 s	One leg + clock reach	Heel-to-toe walk 2 fwd + backward walk 2, hand on counter	One leg on a folded towel, at counter	Cloud hands with side steps 2 min	All of the above, 2 rounds
Walking (most days)	45 min brisk	45 min, hills	50 min, 5×2 min fast	50 min, hills	60 min	60 min, hills

Monthly Strength Age retest

Start, then end of weeks 4, 8 and 12. Same chair, shoes, time of day and curl weight. Instructions: the 7-Day Strength Reset or the app. Fitness checks from senior fitness research (E49; CDC STEADI E11, E50, E51), **not a medical test**.

Name: _____ Age: _____ Woman / Man

Test	Start: date _____	Week 4: _____	Week 8: _____	Week 12: _____
1. Chair stands in 30 s (write H if hands used)				
2. Balance: last stage held 10 s (0-4)				
3. Arm curls in 30 s (what I lifted: ____)				
4. Step test: right-knee lifts in 2 min				
5. Sit-and-reach, inches (+/-)				
6. Timed Up and Go, seconds (12+ = tell your doctor)				
Chair points = $2.5 \times (\text{midpoint} - \text{stands})$, -12 to +12; hands used: at least +8				
Balance points (table on the next page)				
My Strength Age = age + chair points + balance points				

Your Strength Age chart

Put a dot for each retest and join them. It's a motivational estimate from published norms; the app version adds a few questions, so it can differ by a year or two.

Write ages here				
	Start	Week 4	Week 8	Week 12

The numbers you need

Age	Women midpoint	Men midpoint	Women typical	Men typical
60–64	14.5	16.5	12–17	14–19
65–69	13.5	15.0	11–16	12–18
70–74	12.5	14.5	10–15	12–17
75–79	12.5	14.0	10–15	11–17
80–84	11.5	12.5	9–14	10–15
85–89	10.5	11.0	8–13	8–14
90–94	7.5	9.5	4–11	7–12

Last balance position held 10 s	Points
Couldn't hold position 1	+10
Held position 1 only (feet together)	+8
Held position 2 (half step)	+5
Held position 3 (heel to toe)	0 (age 80+: –2)
Held position 4 (one foot)	–3 (age 80+: –5)

Under 60? Use 60–64. Strength Age stays within 15 years below and 20 years above your age, and never below 40. "Typical" chair stands = middle 50% for your age (E49). Below the CDC cutoffs, or a Timed Up and Go of 12 seconds or more, is worth mentioning to your doctor (E11, E51).

Habit tracker: 12 weeks

Tick the day when you did the session. At the end of each week, write how many days you did each habit (0–7). Grace rule: a missed day is just a missed day. Don't double up; do tomorrow's session tomorrow.

Week	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Protein breakfast (days)	Walk after a meal (days)	Toothbrush balance (days)
Wk 1	☐	☐	☐	☐	☐	☐	☐			
Wk 2	☐	☐	☐	☐	☐	☐	☐			
Wk 3	☐	☐	☐	☐	☐	☐	☐			
Wk 4	☐	☐	☐	☐	☐	☐	☐			
Wk 5	☐	☐	☐	☐	☐	☐	☐			
Wk 6	☐	☐	☐	☐	☐	☐	☐			
Wk 7	☐	☐	☐	☐	☐	☐	☐			
Wk 8	☐	☐	☐	☐	☐	☐	☐			
Wk 9	☐	☐	☐	☐	☐	☐	☐			
Wk 10	☐	☐	☐	☐	☐	☐	☐			
Wk 11	☐	☐	☐	☐	☐	☐	☐			
Wk 12	☐	☐	☐	☐	☐	☐	☐			

The habits: 25–30 g protein at breakfast (kidney disease: ask your doctor for your number, E28) · a few minutes of walking after your biggest meal (E23) · heel-to-toe while you brush your teeth, hand on the sink (E13) · and call someone this week (E37).

Where the numbers come from

Every number and health statement in this book traces to one of the entries below. They are the same evidence IDs our whole team uses (EVIDENCE.md). Grade A means a Cochrane review, a large meta-analysis or a guideline body. B means a single good trial or a small meta-analysis. C means an observational study: it shows a link, not cause and effect. D means lab, physiology or label data.

E01 (grade A). Progressive resistance training in older adults: 121 RCTs, 6,700 people, 2–3×/week. Large effect on strength (SMD 0.84), moderate-to-large effect on getting out of a chair (SMD –0.94), gait speed +0.08 m/s, less osteoarthritis pain. Most side effects were minor soreness or joint pain.

Source: Liu & Latham, Cochrane 2009 — https://www.cochrane.org/evidence/CD002759_progressive-resistance-strength-training-improving-physical-function-older-adults

E02 (grade A). NSCA position statement: resistance training is safe and effective for older adults, including frail people. Progress gradually. Train 2–3×/week, 1–3 sets of 6–12 reps at 70–85% 1RM once adapted. Include power (fast lifting intent) and balance.

Source: Fragala et al., J Strength Cond Res 2019 — https://journals.lww.com/nsca-jscr/fulltext/2019/08000/resistance_training_for_older_adults__position.1.aspx

E11 (grade A (tool)). CDC STEADI 30-second chair stand (17-inch chair, arms crossed), used in the CDC's older-adult screening. Below-average scores: men 60–64 <14, 65–69 <12, 70–74 <12, 75–79 <11, 80–84 <10, 85–89 <8, 90–94 <7. Women 60–64 <12, 65–69 <11, 70–74 <10, 75–79 <10, 80–84 <9, 85–89 <8, 90–94 <4. If arms are used to stand, the score is 0.

Source: CDC STEADI — <https://www.cdc.gov/steady/media/pdfs/STEADI-Assessment-30Sec-508.pdf>

E13 (grade A–). Otago Exercise Programme: a home programme of leg strengthening, balance exercises and walking designed for older adults, studied in 7 trials with 1,503 people (mean age 81.6). We use its exercise types and progressions; we make no outcome claim from it.

Source: Thomas et al., Age Ageing 2010 — <https://www.ncbi.nlm.nih.gov/books/NBK78898/>

E23 (grade A– (meta)). Light walking after meals (even 2–5 minutes) lowers the post-meal glucose rise compared with sitting.

Source: Buffey et al., Sports Med 2022 — <https://pubmed.ncbi.nlm.nih.gov/35147898/>

E28 (grade A (consensus)). PROT-AGE: adults over 65 should get 1.0–1.2 g protein/kg/day, and 1.2–1.5 g/kg if active or ill (unless there is severe kidney disease). Aim for 25–30 g per meal with about 2.5–2.8 g leucine.

Source: Bauer et al., JAMDA 2013 — <https://pubmed.ncbi.nlm.nih.gov/23867520/>

E37 (grade C (meta of cohorts)). 148 studies, 308,849 people. Stronger social relationships were linked with a 50% higher likelihood of survival (OR 1.50), comparable to well-known risk factors like smoking.

Source: Holt-Lunstad et al., PLoS Med 2010 — <https://journals.plos.org/plosmedicine/article?id=10.1371%2Fjournal.pmed.1000316>

E41 (grade B (known literature)). Pain-monitoring model: during and after exercise, pain up to about 3/10 (we use a conservative threshold) is acceptable if it settles by the next morning and isn't trending up week to week.

Source: Thomeé, Phys Ther 1997; Silbernagel et al., Am J Sports Med 2007

E42 (grade A (statement)). AHA scientific statement (2023): resistance training is recommended for adults with and without cardiovascular disease. It lowers blood pressure and improves glucose control and function. Avoid breath-holding (Valsalva). People with CVD should get individualized clearance.

Source: Paluch et al., Circulation 2024 — <https://www.ahajournals.org/doi/10.1161/CIR.0000000000001189>

E43 (grade A (guideline)). Pre-exercise screening (ACSM 2015): anyone with symptoms of cardiovascular, metabolic or renal disease (chest pain, unusual breathlessness, fainting) should get medical clearance before starting. Otherwise, light-to-moderate exercise can begin and progress.

Source: Riebe et al., Med Sci Sports Exerc 2015 (known literature)

E49 (grade A (tool/norms)). Senior Fitness Test (Rikli & Jones): 7,183 US community-dwelling adults aged 60–94 from 267 sites in 21 states. "Normal range" = the middle 50% (25th–75th percentile) for each 5-year age band and sex. 30-s chair stand, 30-s arm curl (5 lb women / 8 lb men), 2-minute step test (right-knee lifts to midway between kneecap and hip bone), chair sit-and-reach (inches, + past the toes / – short of them). Example normal ranges, women 70–74: chair stand 10–15, arm curl 12–17, step test 68–101, sit-and-reach –1.0 to +4.0 in; men 70–74: 12–17, 14–21, 80–110, –3.5 to +2.5 in. Full tables in products/strength_reset.md. It's a fitness comparison, not a diagnosis.

Source: Rikli & Jones, Senior Fitness Test Manual, 2nd ed. 2013 (primary not opened). Opened: normative tables — <https://integrativephysicaltherapyservices.com/wp-content/uploads/2023/08/Fitness-Testing-for-Seniors-PDF.pdf> ; protocol + sample size — <https://fitnessnorms.com/functional/chair-sit-and-reach/> , <https://fitnessnorms.com/functional/step-test/> , <https://fitnessnorms.com/functional/arm-curl/> . Note: the PDF's women 60–64 sit-and-reach cell reads "–0.5 to +0.5" (typo); we use the fitnessnorms 25th–75th percentiles (–0.6 to +4.8 in).

E50 (grade A (tool)). CDC STEADI 4-Stage Balance Test: feet side by side, semi-tandem (instep beside big toe), tandem (heel to toe), one leg; up to 10 s each, stop at the first stage not held. The CDC uses the tandem stand as a screening checkpoint: under 10 s is worth discussing with a clinician.

Source: CDC STEADI — opened <https://www.cdc.gov/steady/media/pdfs/STEADI-Assessment-4Stage-508.pdf>

E51 (grade A (tool)). CDC STEADI Timed Up and Go: stand from a chair, walk 10 feet (3 m) at normal pace, turn, walk back, sit. The CDC uses 12 seconds or more as a screening threshold worth discussing with a clinician. Always have someone stand by.

Source: CDC STEADI — opened <https://www.cdc.gov/steady/media/pdfs/STEADI-Assessment-TUG-508.pdf>